

13.2 Derivatives and Integrals of Vector Functions

1 Warm Up

Question 1: What is the definition of the derivative?

Problem 1: Find the derivative of $\cos x$, $\ln x$, xe^x , $(1 + \sqrt{x})^3$

2 Motivation

We have defined vector functions which seem useful to describe motion and curves/surfaces in 3-space. It seems natural to wonder if we have a notion of the derivative, which also helps us analyze the motion of functions?

3 Definitions & Theorems

1. If r is a vector function, the derivative of r denoted r' is defined the same way we define it in calc I.

$$r'(t) = \lim_{h \rightarrow 0} \frac{r(t+h) - r(t)}{h}.$$

If this limit exists, geometrically it represents the **tangent vector** to the curve defined by r . It is a direction vector to the line tangent to the curve at a given t .

2. This tangent vector will be useful to know and thus we define a **unit tangent vector** $T(t)$ as follows.

$$T(t) = \frac{r'(t)}{|r'(t)|}.$$

3. To compute the derivative for these vector functions we utilize the following theorem.

Theorem 1. If $r(t) = \langle f(t), g(t), h(t) \rangle$ where f, g, h are differentiable functions then $r'(t) = \langle f'(t), g'(t), h'(t) \rangle$.

4. Many of the rules when taking the derivatives of functions also hold when taking the derivative of vector functions. These include the following.

- $(u(t) + v(t))' = u'(t) + v'(t)$
- $(cu(t))' = cu'(t)$

- $(f(t)u(t))' = f'(t)u(t) + f(t)u'(t)$
 - $(u(t) \cdot v(t))' = u(t) \cdot v'(t) + u'(t) \cdot v(t)$
 - $(u(t) \times v(t))' = u'(t) \times v(t) + u(t) \times v'(t)$
 - $(u(f(t)))' = f'(t)u'(f(t))$
5. If we can find the derivative of a vector function which gives us a direction vector which is tangent to our space curve, we can use that, along with a point (Point of tangency) to find the **equation of the tangent line**! Just like we did in 12.5.
6. Similarly we can find the **integral** of a vector functions as follows.

$$\int_a^b r(t) = \left(\int_a^b f(t) \right) i + \left(\int_a^b g(t) \right) j + \left(\int_a^b h(t) \right) k.$$

4 Examples

Problem 1. Find $r'(t)$ if $r(t) = \langle 1/t, 1/t^2, 1/t^3 \rangle$.

Problem 2. Find $r'(t)$ if $r(t) = e^t \sin(t)i - 2^t j + (1 + \ln t)^3 k$.

Problem 3. Find $r''(t)$ in problem 1.

Problem 4. Find $\int r(t)$ where $r(t) = \sqrt{t}i + 2t/(t^2 + 5)j - 4k$.

Problem 5. If $u(t) = \langle 3t^2, \sqrt[3]{t}, -5t \rangle$ and $v(t) = \langle 2t + 3, -t^2, t - t^2 \rangle$, find $(u(t) \cdot v(t))'$ and $(u(t) \times v(t))'$.

5 Scratch Work